

students, this is a good field to start their career. What they have studied during their engineering course, they should apply to circuit design and PCB design. Initially, engineering graduates can enter into this field as a circuit designer or a board designer."

A glimpse

PCB designers are key persons in research and development (R&D), electronics production units as well as in the PCB industry. A PCB designer's job comes with a lot of responsibilities. "Nowadays it's not only about making interconnections between electronic components but also understanding the functioning of the circuit. It is often observed that a circuit which functions well when assembled over a bread board may or may not function properly over a poorly designed PCB layout. Depending upon the complexity of the circuit or PCB it may take two to three days for system engineers to locate and rectify the problems," Singla says.

There can be numerous reasons for such kind of failures; PCB design and fabrication being one of them. "If a PCB is not designed or fabricated as per the circuit requirements, the circuit may not function at all, or function partially. Generally, circuit designers overlook the PCB design/fabrication part so as to reduce the product's time-to-market or to compensate for the delay. Sometimes PCB designers or fabricators are not allowed to devote sufficient time as intended," Singla explains.

According to Vikas Kohli, senior architect, Cadence Design Systems India, "New computing, network, and telecommunication products have in-

Without PCB, we can not expect any electronic circuit. Since a PCB provides physical support as well as interconnections between different electronic components, there exist great opportunities as well as very good growth in this field worldwide — Shavinder Singla, deputy engineer in PCB Lab at Centre for Development of Advanced Computing (CDAC), Mohali



radically altered the PCB design landscape. A recent report titled 'Design Trends and EDA Tool Usage—Asia Pacific,' highlights consumer electronics, telecom/data communications and industrial control/medical electronics as dominant industries in the region. The PCB design section report shows that over 50 per cent of designs have less than 50 integrated circuits (ICs) on a board. In fact, 20 per cent of designs have less than four ICs on the board. The pin-count of these devices has increased, so has the complexity of the logical interconnects. In this case, a smaller number of larger components means a larger number of design headaches."

"One design discipline greatly affected by this change is schematic design capture. To keep pace with shrinking market windows in light of all this complexity, designers have to capture designs faster than ever before. And to meet profitability goals they have to design it right the first time," Kohli adds.

What your pocket says

Let's turn to money matters. This field offers enough provisions for learning as well as earning. The starting salary for a fresher is typically between Rs

150,000 and Rs 200,000 per annum with impressive year to year hikes. Coming to experienced professionals, the PCB industry offers salary in the range of Rs 500,000 to Rs 600,000 per annum for people with four to five years of experience.

"As a fresher one can not expect high salary. But after an experience of five-six years, and with understanding and implementation of standards, one can expect a very good package and may act as a team leader," adds Singla. And of course if you grab an international opportunity, the pay is quite good even for freshers. It is needless to mention that PCB designers can experience much better growth while working with an MNC.

"It's been my experience that the designers who work on the higher-end or more expensive CAD tools (i.e., Cadence Allegro and Mentor Graphics) seem to get paid a little higher than designers who work on the lower-end or less expensive systems like Altium and PADS. As I mentioned earlier, back in the late 90s, as long as you were a good designer you would be hired regardless of the CAD tool. But now, unfortunately, hiring-managers seem to put an emphasis on which CAD systems you're running and even what version, etc," opines Derek Doucette, an executive with American Contract Group, a Massachusetts-based staffing firm. He has been placing candidates into PCB design positions since the 1990s.

"A designer once told me, "Besides a few features here and there, they basically all do the same thing.

For electricals and electronics engineering (EEE) and electronics communication engineering (EC) students, this is a good field to start their career — K. Krithiga, technical lead, CADD Center

